

Name:	
ID:	
Date:	

Bachelor of Science in Electrical Engineering Worksheet

Electrical Engineering (EE) Requirements (47 credits minimum)

Electrical Engineering Core (24 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	EE 201	Linear Circuit Analysis I	3	Pre: ENGR 131, MA 166 (min grade of C-), PHYS 172 ConP: MA 261	Lect.: 3h			
☐	EE 202	Linear Circuit Analysis II	3	Pre: EE 201 (min grade of C) ConP: MA 266	Lect.: 3h			
☐	EE 207	Electronic Measurement Techniques Lab	1	ConP: EE 201	Lab.: 3h			
☐	EE 208	Electronic Devices and Design Laboratory	1	Pre: EE 207 ConP: EE 255	Lab.: 3h			
☐	EE 255	Introduction to Electronic Analysis and Design	3	Pre: EE 201 (min grade of C)	Lect.: 3h			
☐	EE 270	Introduction to Digital System Design	4	ConP: EE 201	Lect.:3h; Lab.: 3h			
☐	EE 301	Signals and Systems	3	Pre: EE 202 (min grade of C), MA 266	Lect.: 3h			
☐	EE 302	Probabilistic Methods in Electrical and Computer Engineering	3	Pre: MA 266 ConP: EE 301	Lect.: 3h			
☐	EE 311	Electric and Magnetic Fields	3	Pre: EE 201, PHYS 272, MA 266	Lect.: 3h			

Name:	
ID:	
Date:	

Electrical Engineering Seminars (1 credit)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	EE 200	Electrical and Computer Engineering Sophomore Seminar	0	Sophomore	Lect.: 1h			
☐	EE 400	Professional Development and Career Guidance - Graduation Project I	1	Pre: EE 200, Senior	Lect.:1h; Lab.: 1h			

Senior Design Requirement (3-4 credit)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	EE 402	Electrical Engineering Design Projects	3	Pre: EE 400, EE Core Curriculum, Senior	Lect.:1h; Lab.: 1h			
☐	CE 477	Digital Systems Senior Project	4	Pre: CE 400 or EE 400, EE Core Curriculum or CE Core Curriculum, Senior	Lect.:1h; Lab.: 1h			

Advanced Electrical Engineering Selectives (9-11 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	EE 305	Semiconductor Devices	3	Pre: EE 255, PHYS 272, MA 266	Lect.: 3h			
☐	EE 321	Electromechanical Motion Devices	3	Pre: EE 202, PHYS 272 ConP: EE 255	Lect.: 3h			
☐	EE 382	Feedback System Analysis and Design	3	Pre: EE 301	Lect.: 3h			
☐	EE 438	Digital Signal Processing with Applications	4	Pre: EE 301, EE 208, EE 302	Lect.:3h; Lab.: 3h			
☐	EE 440	Transmission of Information	4	Pre: EE 301, EE 208, EE 302	Lect.:3h; Lab.: 3h			
☐	CE 362	Microprocessor Systems and Interfacing	4	Pre: EE 270 (min grade of C) or CE 270 (min grade of C), CS 159	Lect.:3h; Lab.: 3h			

Name:	
ID:	
Date:	

Electrical Engineering Electives (7-10 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐								
☐								
☐								
☐								

Electrical Engineering Laboratory Requirement: Three (3) Electrical Engineering Upper-Level Laboratory courses or Electrical Engineering courses with laboratory components in addition to those required as part of the Electrical Engineering Core Curriculum (EE 207, 208, and 270). Courses with laboratory components taken as Advanced Electrical Engineering Electives, EE 362, 438 and 440, also contribute to this requirement.

Electrical Engineering Credits Planned: _____
 Electrical Engineering Credits Completed: _____
 Electrical Engineering Credits Remaining: _____

General Engineering (10 credits)

Introduction to Engineering (7 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	CS 159	Programming Applications for Engineers	3	Pre: ENGR 131	Lect.: 3h; Lab.: 2h			

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	ENGR 131	Transforming Ideas to Innovation I	2	-	Lect.:2h; Lab.: 2h			
☐	ENGR 132	Transforming Ideas to Innovation II	2	Pre: ENGR 131	Lect.:2h; Lab.: 2h			

OR

☐	ENGR 100	First-Year Engineering Lectures	1	-	Lect.: 1h			
☐	ENGR 126	Engineering Problem Solving and Computer Tools	3	-	Lect.: 3h			

Name:	
ID:	
Date:	

Engineering Breadth Requirement (3 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	ENGR 200	Thermodynamics I	3	Pre: CHM 115, ConP: MA 261, ENGR 132	Lect.: 3h			
☐	ENGR 297	Basic Mechanics I (Statics)	3	Pre: PHYS 172, ConP: MA 261	Lect.: 3h			
☐	IE 335	Operation Research Optimization	3	Pre: MA 265, ConP: EE 302 or CE 302 or IE 332	Lect.: 2h; Lab.: 2h			
☐	IE 336	Operation Research Stochastic Models	3	Pre: MA 265, IE 230 ConP: EE 302 or CE 302 or IE 332, MA 266	Lect.: 3h			
☐	MSE 230	Structure and Properties of Materials	3	Pre: CHM 115 (min. grade C-), MA 165 (min grade C-)	Lect.: 3h			
☐	NUCL 200	Introduction to Nuclear Engineering	3	Pre: MA 166 or equivalent, PHYS 172 or equivalent	Lect.: 3h			
☐	NANO 101	Introduction to Nanotechnology	3	Pre: MA 261; MA 266 or MA 262; PHYS 272 or PHYS 241; CHM 115	Lect.: 3h			
☐	CVL 355	Engineering Environmental Sustainability I	3	Pre: Sophomore	Lect.: 3h			

General Engineering Credits Planned: _____

General Engineering Credits Completed: _____

General Engineering Credits Remaining: _____

Name:	
ID:	
Date:	

Mathematics Requirement (18 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	MA 165	Analytic Geometry and Calculus I	4	Passing Math Placement Test or MAT 110 or MA 158	Lect.:3h; Lab.: 2h			
☐	MA 166	Analytic Geometry and Calculus II	4	Pre: MA 165	Lect.: 4h			
☐	MA 261	Multivariate Calculus	4	Pre: MA 166	Lect.: 4h			
☐	MA 265	Linear Algebra	3	Pre: MA 166	Lect.: 3h			
☐	MA 266	Ordinary Differential Equations	3	Pre: MA 261	Lect.: 3h			

Mathematics Credits Planned: _____

Mathematics Credits Completed: _____

Mathematics Credits Remaining: _____

Science Requirement (15-16 credits)

Core (12 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	CHM 115	General Chemistry I	4	ConP: MA 165	Lect.:3h; Lab.: 2h			
☐	PHYS 172	Modern Mechanics	4	ConP: MA 165	Lect.:3h; Lab.: 2h			
☐	PHYS 272	Electric and Magnetic Interactions	4	Pre: PHYS 172, ConP: MA 166	Lect.:3h; Lab.: 2h			

Science Electives (3-4 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	BIOL 110	Fundamentals of Biology I	4	-	Lect.:3h; Lab.: 2h			
☐	CHM 116	General Chemistry II	4	Pre: CHM 115	Lect.:3h; Lab.: 2h			

Name:	
ID:	
Date:	

☐	PHYS 322	Intermediate Optics	3	Pre: PHYS 272 or PHYS 241	Lect.: 3h			
☐	PHYS 342	Modern Physics	3	Pre: PHYS 272 or PHYS 241	Lect.: 3h			
☐	(BIOL 121	Biology I: Diversity, Ecology, and Behavior	2	-	Lect.:2			
☐	and BIOL 135)	First year Biology Laboratory	2	ConP: BIOL 121, CHM 115	Lab.: 3			
☐	CHM 112	General Chemistry	3	Pre: CHM 115	Lect.: 3h			
☐	EAPS 125	Environmental Science and Conservation	3	-	Lect.: 3h			

Science Credits Planned: _____

Science Credits Completed: _____

Science Credits Remaining: _____

Liberal Arts Requirements (28 credits)

English Language and Communication Skills (10 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	COM 114	Fundamentals of Speech Communication	3	Pre: ENGL 100	Lect.: 3h			
☐	ENGL 100	English for Academic Studies	3	-	Lect.: 3h			
☐	ENGL 106	First-Year Composition	4	Pre: ENGL 100	Lect.: 4h			

General Education Electives (18 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐								
☐								
☐								

Name:	
ID:	
Date:	

Liberal Arts Credits Planned: _____

Liberal Arts Credits Completed: _____

Liberal Arts Credits Remaining: _____

Complementary Electives (up to 13 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐								
☐								
☐								
☐								

Complementary Electives Credits Planned: _____

Complementary Electives Credits Completed: _____

Complementary Electives Credits Remaining: _____

Minimum Total Credits Required for Degree: 128

Total Credits Planned: _____

Total Credits Completed: _____

Total Credits Remaining: _____